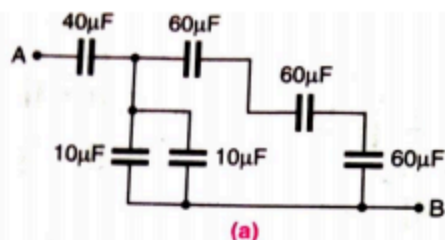


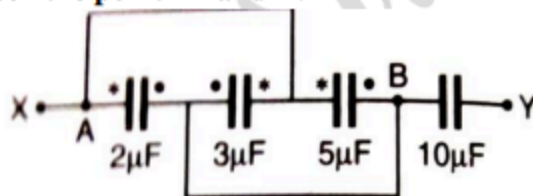
NUMERICAL PROBLEMS IN CAPACITORS

MANDATORY PROBLEMS

1. Two capacitors of capacitances $6\ \mu\text{F}$ and $12\ \mu\text{F}$ are connected in series with a battery. The voltage across the $6\ \mu\text{F}$ capacitors is 2V . Compute the total battery voltage.
2. Find the equivalent capacitance of the combination of capacitors shown in Figure. Also, calculate the total charge that flows in the circuit when 100 V battery is connected between the points A and B.



3. Four capacitors are connected as shown in Fig. Calculate the equivalent capacitance between the point X and Y.

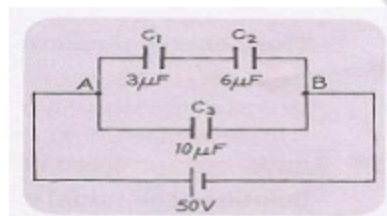


4. A $4\ \mu\text{F}$ capacitor is initially isolated and has a potential difference of 10 V across its plates. The plates of this capacitor are then connected in parallel to the plates of an initially uncharged $12\ \mu\text{F}$ capacitor. Find the electrical potential energy stored before and after connection. Comment upon the reason for this change in potential energy.
5. Keeping the voltage of the charging source constant, what would be the percentage change in the energy stored in a parallel-plate capacitor if the separation between its plates were to be decreased by 10% ?
6. A parallel plate air capacitor has a capacitance of $5\ \mu\text{F}$. It becomes $50\ \mu\text{F}$ when a dielectric medium occupies the entire space between its two plates. What is the dielectric constant of the medium?
7. You are provided with $8\ \mu\text{F}$ capacitors. Show with the help of a diagram how you will arrange minimum number of them to get a resultant capacitance of $20\ \mu\text{F}$.
8. A parallel plate capacitor of plate area $A = 600\text{ cm}^2$ and plate separation $d = 2.0\text{ mm}$ is connected to a DC source of 200 V . Calculate in SI unit: (i) the magnitude of the uniform electric field E between the plates, (ii) the charge density σ on any plate.

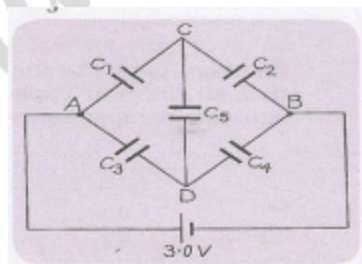
9. An isolated $16 \mu\text{F}$ parallel plate air capacitor has a potential difference of 100 V (fig). A dielectric slab having relative permittivity $= 5$ is introduced to fill the space between the two plates completely. Calculate: (i) the new capacitance of the capacitor (ii) the new potential difference between the two plates of capacitor.



11. Three capacitors $C_1 = 3 \mu\text{F}$, $C_2 = 6 \mu\text{F}$ and $C_3 = 10 \mu\text{F}$ are connected to a 50 V battery as shown in figure. Calculate (i) the equivalent capacitance of the circuit between points A & B, (ii) the charge on C_1 .

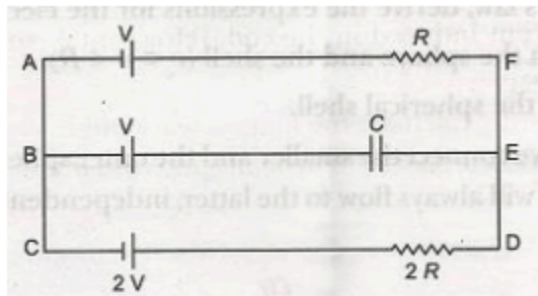


12. Find the equivalent capacitance of the adjoining circuit between the junctions A and B; given $C_1 = C_2 = C_3 = C_4 = 10 \mu\text{F}$ and $C_5 = 5 \mu\text{F}$.



13. Three capacitors each of capacitance C are connected in series. Their equivalent capacitance is C_s . The same three capacitors are now connected in parallel. Their equivalent capacitance becomes C_p . Find the ratio (C_p/C_s) .

14. In the given circuit in the steady state, obtain the expressions for (a) the potential drop (b) the charge and (c) the energy stored in the capacitance C .

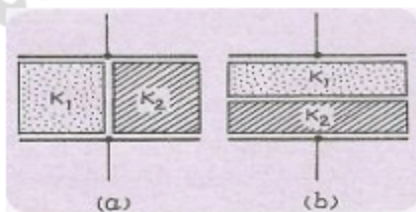


15. A parallel plate capacitor is charged by a battery, which is then disconnected. A dielectric slab having dielectric constant (relative permittivity) K , is now introduced between its two plates in order to occupy the space completely. State in terms of K , its effect on the following:

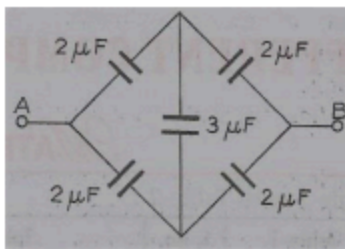
- (i) the capacitance of the capacitor
- (ii) the potential difference between its plates
- (iii) the energy stored in the capacitor.

PRACTICE PROBLEMS

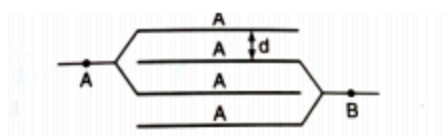
1. What is the area of the plates of a 2F parallel-plate capacitor, given that the separation between the plates is 0.5 cm ?
2. Three capacitors, each of $3\text{ }\mu\text{F}$ capacitance, are connected together in series and are also connected in series with three other similar capacitors which are grouped together in parallel. Sketch the arrangement and calculate its total combined capacitance.
3. 3 capacitors of capacitances 2 pF , 3 pF & 4 pF are connected in parallel.
(i) What is the total capacitance (ii) Determine the charge on each capacitor if the combination is connected to 100 V supply.
4. The plates of a parallel-plate capacitor have an area of 90 cm^2 each and are separated by 2.5 mm . The capacitor is charged by connecting it to a 400 V supply.
(a) How much electrostatic energy is stored by the capacitor?
(b) View this energy as stored in the electrostatic field between the plates and obtain the energy per unit volume, u_E . Hence, arrive at a relation between u_E and magnitude of electric field E between the plates.
5. 125 identical drops of a liquid each of capacitance $5\text{ }\mu\text{F}$ coalesce to form a big drop. Find the capacitance of the big drop.
6. On charging a parallel-plate capacitor to a potential V , the spacing between the plates is halved and a dielectric medium of $\epsilon_r = 10$ is introduced between the plates, without disconnecting the dc source. Explain using suitable expressions, how the:
(i) capacitance, (ii) electric field and (iii) energy density of the capacitor change.
7. The effective capacitance of 2 capacitors connected in series and parallel are $2.4\text{ }\mu\text{F}$ and $10\text{ }\mu\text{F}$ respectively. Calculate the individual capacitances.
8. A parallel-plate capacitor is filled with two dielectrics as shown in Figure. What is its capacitance?



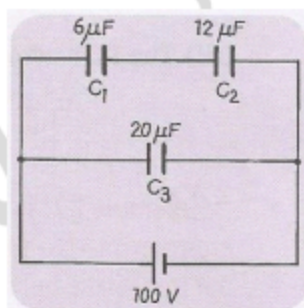
9. Find the equivalent capacitance between the points A and B in the given circuit.



10. Two parallel plate capacitors, X and Y (joined in series with a 12V supply), have the same area of plates and same separation between them. X has air between the plates while Y contains a dielectric of $\epsilon_r = 4$. (i) Calculate capacitance of each capacitor if the equivalent capacitance of the combination is $4 \mu\text{F}$. (ii) Calculate the pd between the plates of X and Y. (iii) What is the ratio of electrostatic energy stored in X and Y?
11. What is the capacitance of arrangement of 4 plates, each of area A, at a distance d in air as shown in Figure?



12. A 600 pF capacitor is charged by a 200 V supply. It is then disconnected from the supply and is connected to another uncharged 600 pF capacitor. How much electrostatic energy is lost in the process?
13. Three capacitors $C_1 = 6 \mu\text{F}$, $C_2 = 12 \mu\text{F}$ and $C_3 = 20 \mu\text{F}$ are connected to a 100 V battery as shown in figure. Calculate: (i) charge on each plate of capacitor C_1
(ii) electrostatic potential energy stored in capacitor C_3 .



14. Two capacitors of unknown capacitances C_1 and C_2 are connected first in series and then in parallel across a battery of 100 V . If the energy stored in the two combinations is 0.045 J and 0.25 J respectively, determine the value of C_1 and C_2 . Also calculate the charge on each capacitor in parallel combination.
15. Two capacitors A and B are connected in series with a 100 V battery and it is observed that potential difference across them are 40 V and 60 V respectively. A capacitor of capacitance $6 \mu\text{F}$ is now connected in parallel with B and as a result, the potential difference across B falls to 50 V . Find the capacitances of A and B.